FCDS at AlexandriaX-2026: Improving Context Aware Dialogue English To Dialectal Arabic Machine Translation Via Context Strategies And Quality Aware Decoding

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Abstract

Translating English dialogue into dialectal Arabic depends on the target dialect, conversational context and speaker attributes, and these challenges are compounded by the scarcity of dialectal parallel data. We present our system for the AlexandriaX-2026 Context-Aware English-to-Dialectal Arabic Dialogue Translation constrained track, which restricts participants to the provided Alexandria dataset and models of $\leq$5B parameters. We fine-tune open LLMs, exploring different context composition strategies for training and inference, and find that mismatched strategies can outperform matched ones depending on the model. We further apply quality-aware decoding (QAD) via Minimum Bayes Risk, tuning the utility function and candidate pool size for consistent gains over greedy decoding. Our final system, a fine-tuned Gemma-3-4B-it with spBLEU-based QAD and a pool size of 50 samples, achieves 26.48 spBLEU and 42.3 chrF++, ranking third on the constrained-track leaderboard.

1 Introduction

Dialectal Arabic (DA) encompasses the diverse varieties used in everyday communication across the Arab world (Talafha et al., 2024, 2025). Unlike Modern Standard Arabic (MSA), these varieties exhibit substantial regional variation in orthography, vocabulary, morphology and syntax (Kchaou et al., 2020). The scarcity of high-quality dialectal parallel corpora further limits the ability of machine translation (MT) systems to learn dialect-specific translation patterns (Abdul-Mageed et al., 2024; Pokrywka et al., 2025).

Beyond dialectal variation and resource scarcity, dialogue translation into DA is sensitive to conversational context. Appropriate choices may depend not only on the source sentence, but also on preceding dialogue turns, the target dialect, and sociolinguistic attributes of the speakers. Consequently, translating each utterance independently can lead to outputs that are semantically correct but inappropriate for the intended conversation (Liang et al., 2021; Honda et al., 2023).

The AlexandriaX-2026 Shared Task (El Mekki et al., 2026) addresses these challenges through its Context-Aware English-to-Dialectal Arabic Dialogue Translation sub-task. The task requires systems to translate English dialogue turns into a specified Arabic dialect while conditioning on available dialogue history, target dialect information, speaker and addressee gender, and persona attributes.

Our contribution in the constrained track of this sub-task involves fine-tuning several LLMs, whose size is 5B parameters or less on the provided training data: the Alexandria dataset (EL Mekki et al., 2026). We train our models using different context sizes and strategies, detailed in Section 4.1. Additionally, we apply quality-aware decoding, which we describe in Section 4.2, during inference to further improve the quality of the translations.

We find that the effect of context strategy and size varies depending on the model. Additionally, for one asymmetric configuration tested, where context strategy varies between training inference, performance can sometimes improve relative to symmetric setups across both phases. Our system achieves competitive performance ranking third on the final leaderboard of the constrained track with an average spBLEU of 26.48 and and average chrF++ of 42.3. We make our code publicly available on GitHub[1] for reproducibility.

2 Background

2.1 Related Work

Dialectal Arabic Machine Translation: Dialectal Arabic machine translation has been mainly constrained by the limited availability of parallel corpora (Zbib et al., 2012; Sajjad et al., 2020).

[1] https://github.com/RanwaKhaled/AlexandriaX

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 848–856
October 28-29, 2026 ©2026 Association for Computational Linguistics

Prior work has tried addressing these issues by constructing parallel dialectal datasets for training and evaluation of MT systems (Bouamor et al., 2018; Kchaou et al., 2020; Abid, 2020; Sajjad et al., 2020; EL Mekki et al., 2026). Moreover, regional variation across dialects further complicates this task as models struggle to learn a unified representation that captures the dialects' linguistic variety, especially while a number of dialects still remains underrepresented in the available corpora (Abdul-Mageed et al., 2024; Mousi et al., 2025).

Context-Aware Machine Translation: Prior work has shown that adding context and moving beyond utterance-level improves translations' quality, especially in dialogue MT (Maruf and Haffari, 2018; Liang et al., 2021). It was also found that context size and composition (Vincent et al., 2023; Choudhary et al., 2025) along with speaker and scene information (Vanmassenhove et al., 2018; Honda et al., 2023), lead to different translation outcomes.

We extend this line of work by leveraging Alexandria (EL Mekki et al., 2026), an English–Arabic dialectal dialogue parallel corpus, using its dialogue context and metadata tags to improve translation output for the AlexandriaX shared task (El Mekki et al., 2026).

2.2 AlexandriaX Shared Task

The AlexandriaX-2026 Shared Task (El Mekki et al., 2026) targets the development and evaluation of dialectal Arabic MT systems. The shared task comprises of three subtasks: Context-Aware English-to-Dialectal Arabic Dialogue Translation, Cross-Dialect Arabic Machine Translation and Dialectal Arabic MT Error Detection and Classification. We participate in the constrained track of Subtask 1, where we're only allowed to use the Alexandria Dataset (EL Mekki et al., 2026) and models of 5B parameters or less.

3 Data

We conduct our experiments on the Alexandria dataset (EL Mekki et al., 2026), a dialectal Arabic dialogue dataset covering 13 Arabic varieties and 11 mainstream domains. The dataset also provides contextual and sociolinguistic information, including dialect, country, domain, speaker and addressee gender, and persona information. Table 1 summarizes the distribution across dialects and splits.

4 System

To identify the model with the strongest dialectal capabilities for our task, we start by evaluating 11 Arabic-centric and multilingual LLMs (2B to 5B parameters; see Table 2). From these, we select the five top-scoring models to train using the same context-level prompt used in EL Mekki et al. (2026) with two distinct context strategies which we term **default** and **ideal**, discussed in Section 4.1. Finally, we apply hyperparameter tuning to the leading model alongside quality-aware decoding to determine the optimal setup.

4.1 Context Strategy

According to Choudhary et al. (2025), a context strategy defines the specific composition and structure of contextual information provided alongside a source sentence when prompting an LLM to generate its translation. While the original authors explore both reference and non-reference contexts, this work focuses exclusively on the following non-reference strategies across k preceding sentences:

(a) **Source Only** (`src`): Uses the k preceding source-language sentences as context.
(b) **Hypothesis Only** (`hyp`): Uses the k preceding model-generated target translations as context.
(c) **Source + Hypothesis** (`src+hyp`): Combines the k preceding source sentences with their corresponding model-generated hypotheses.

In EL Mekki et al. (2026) and the organizers' baseline code, the `src+hyp` strategy is adopted across all available previous turns ($k = n - 1$), which we refer to as the **default setup**. However, our preliminary experiments with the baseline `Nile-Chat` model revealed that using the `src` strategy constrained to the two most recent turns ($k = 2$) yields some performance gains. We designate this configuration as our **ideal setup**.

4.2 Quality Aware Decoding

Quality Aware Decoding (QAD) (Mohammed et al., 2026) was used at inference to produce more dialect-faithful translations. Rather than selecting the single highest-probability hypothesis, QAD draws a pool of candidate translations from the model's output distribution and selects the candidate that maximizes expected utility against the rest of the pool, using each remaining candidate as a pseudo-reference.

This procedure approximates the Bayes-optimal hypothesis under a sentence-level utility metric

Split	EG	JO	LB	LY	MA	MR	OM	PS	SA	SD	SY	TN	YE	Total
Train	3,108	5,501	8,906	0	2,573	5,515	6,280	14,933	8,470	0	6,071	2,034	3,089	66,480
Dev	1,113	1,113	1,118	0	1,110	1,114	1,109	1,110	1,110	0	1,119	1,116	1,118	12,250
Public Test	1,118	1,107	1,106	1,109	1,115	1112	1,118	1,109	1,113	1,106	1,114	1,109	1,106	14,442
Private Test	1,113	1,109	1,110	1,309	1,111	1119	1,107	1,111	1,114	915	1,114	1,114	1,113	14,459

Table 1: Distribution of dialogue turns across dialects and dataset splits in the Alexandria dataset. (EG: Egyptian, JO: Jordanian, LB: Lebanese, LY: Libyan, MA: Moroccan, MR: Mauritanian, OM: Omani, PS: Palestinian, SA: Saudi, SD: Sudanese, SY: Syrian, TN: Tunisian, and YE: Yemeni.)

Model	Type
NileChat-3B-Base (El Mekki et al., 2025)	Base
NileChat-3B (El Mekki et al., 2025)	Instruct
Qwen3-4B-Base (Yang et al., 2025)	Base
Qwen3-4B (Yang et al., 2025)	Instruct
Llama-3.2-3B (Grattafiori et al., 2024)	Base
Llama-3.2-3B-Instruct (Grattafiori et al., 2024)	Instruct
Tiny-aya-base (Salamanca et al., 2026)	Base
Tiny-aya-earth (Salamanca et al., 2026)	Instruct
Gemma-3-4B-it (Team et al., 2025)	Instruct
Gemma-4-E4B-it (Team et al., 2026)	Instruct
SILMA-Kashif-2B (SILMA-AI, 2025)	Instruct

Table 2: Models used in our experiments and their types, Base or Instruct.

without requiring access to a reference translation at inference, making it directly applicable under the shared task's held-out test conditions.

QAD is applied independently at each turn and the hypothesis selected by Minimum Bayes Risk (MBR) for the turn t is what is added to the conversation history and used to condition the prompt for the turn $t+1$, so the effect of QAD is compounded in a conversation. The specific choice of the utility and the candidate pool size N are treated as tunable hyperparameters, which we search over empirically in Section 6.4.

5 Experimental Setup

Hyperparameters: Initial experiments across the top five models employed 4-bit QLoRA with an effective batch size of 128 for one epoch using a learning rate of 2×10^{-4}, without applying chat templates. This configuration was fixed across all models to ensure a consistent comparison.

Evaluation: Models were evaluated using the public test split of the Alexandria Dataset, using two metrics: spBLEU (Goyal et al., 2022) as the main metric and chrF++ (Popović, 2017) as an additional metric. To assess cross-configuration robustness, every fine-tuned model was tested using both the **default** and **ideal** prompt strategies during inference.

6 Model Selection

6.1 Candidate Model Screening

We first evaluate all 11 candidate models, with scores shown in Appendix C, without any fine-tuning, under both the **default** and **ideal** context strategies. Based on these zero-shot results, we select the five highest-scoring models: `NileChat-3B-Base`, `NileChat-3B`, `Qwen3-4B-Base`, `Qwen3-4B-Instruct`, and `Gemma-3-4b-it`, for fine-tuning, using the hyperparameter configuration described in Section 5.

6.2 Selecting the Leading Configuration

Table 3 reports the results of fine-tuning the five selected models under both the ideal and default training strategies, each evaluated with both prompt configurations at inference. Across all model-strategy-prompt combinations, `gemma-3-4b-it` trained under the **ideal** strategy and evaluated with the **default** prompt at inference achieves the highest overall scores. Notably, our top configuration is trained with the reduced, source-only ideal context but performs best at inference time when given the full default context. This indicated that the *ideal* strategy adopted from preliminary experiments on the baseline NileChat model (Section 4.1), transfers effectively to some of the evaluated models especially to `gemma-3-4b-it`. As a result, this asymmetric configuration is carried for further hyperparameter tuning.

6.3 Hyperparameter Tuning

Building on the top-performing model from Table 3, `gemma-3-4b-it`, we iteratively refined the training configuration by disabling 4-bit quantization, applying the chat template, increasing the number of training epochs to two, and reducing the effective batch size to 32 to increase gradient updates frequency. Additionally, we introduced a weight decay of 0.05 and increased the LoRA capacity to $R = 32$ and $\alpha = 64$. We also evaluated a reduced

	Ideal Setup (Training)				Default Setup (Training)			
	Default Prompt		Ideal Prompt		Default Prompt		Ideal Prompt	
Model	**spBLEU**	**chrF++**	**spBLEU**	**chrF++**	**spBLEU**	**chrF++**	**spBLEU**	**chrF++**
NileChat-3B-Base	20.99	37.42	**21.85**	**37.87**	20.64	36.91	**21.14**	**37.16**
NileChat-3B	22.80	38.32	**23.34**	**39.24**	22.42	38.56	**22.68**	**38.55**
Qwen3-4B-Base	**18.91**	**35.58**	18.86	35.39	**20.57**	**36.02**	20.13	35.29
Qwen3-4B-Instruct	**19.44**	**35.78**	19.42	35.62	**19.22**	**35.49**	18.89	35.24
Gemma-3-4B-it	**23.83**	**39.71**	23.68	39.48	**23.53**	**39.35**	23.14	38.90

Table 3: Fine-tuning results for the five selected models under the Default and Ideal context strategies, evaluated using both prompt configurations. Bold values indicate the better-performing evaluation prompt for each training strategy. The configuration carried out for further tuning is *Ideal Setup (Training) + Default Prompt*.

#	QAD	Utility	N	spBLEU	chrF++
1	✗	–	–	25.522	41.5758
2	✓	chrF++	16	26.0408	**42.3884**
3	✓	BLEU	32	26.4148	42.2357
4	✓	BLEU	50	**26.4825**	42.3029

Table 4: Leaderboard submissions of our fine-tuned `gemma-3-4b-it` model on the private test split. QAD indicates whether Quality-Aware Decoding was applied, with *Utility* and N denoting the scoring function and number of candidates used (Temperature T is fixed at 0.7 for all QAD runs). Bold marks the best value per metric.

learning rate (from 2×10^{-4} to 1×10^{-4}), which resulted in lower performance. Therefore, the final configuration retained the aforementioned changes while using the original learning rate of 2×10^{-4}. These modifications yielded an increase from 23.83 to 26.64 spBLEU and from 39.71 to 42.23 chrF++.

6.4 QAD Setup Tuning

We tune the two QAD hyperparameters with the largest effect on quality: the utility function and the number of sampled candidates N, using the private test split via our fine-tuned `gemma-3-4b-it` model through the Codabench platform, until reaching the highest spBLEU score.

Starting with chrF++ as the utility function, as it's cheaper and more stable, per Mohammed et al. (2026), with $N = 16$ and temperature $T = 0.7$. We then switch to spBLEU as the utility function since it's our primary metric and scale to $N = 32$, effectively increasing the pool size. This setup yielded roughly $+0.37$ spBLEU at a cost of only -0.15 chrF++, and increasing further to $N = 50$ gave a final marginal gain. We also experimented with raising the sampling temperature, but this consistently degraded performance, so we kept $T = 0.7$ throughout. Full results for each configuration are reported in Table 4.

7 Official Results

Table 4 reports the spBLEU and chrF++ scores for each submission during the test phase. Applying quality aware decoding led to a surge in scores (from 25.52 spBLEU/ 41.57 chrF++ to 26.04 spBLEU/ 42.38 chrF++). Additionally, further tuning of QAD configuration, described in Section 6.4 steadily increased the quality of the translation, with our final submission reaching 26.4825 spBLEU/ 42.3 chrF++. This configuration achieved our best result and ranked third on the constrained-task leaderboard.

8 Discussion

While our system's average scores are competitive, per-dialect performance (Table 5) reveals that training volume alone doesn't account for translation quality. Although dialects with larger training sets often achieve higher spBLEU scores, notable deviations are evident.

To quantify this relationship, we computed both Pearson (Pearson, 1896) ($r = 0.53, p = 0.06$) and Spearman (Spearman, 2010) ($\rho = 0.60, p = 0.03$) correlations between per-dialect training volume and spBLEU, which reflect a moderate positive association between spBLEU and training volume, even if the relationship isn't strictly linear.

Since the correlation results do not fully explain

Dialect	spBLEU	chrF++
EG	29.89	44.35
JO	33.30	48.46
LB	30.21	45.17
LY	21.31	38.90
MA	20.64	36.42
MR	17.26	33.98
OM	26.94	43.35
PS	30.44	45.50
SA	29.97	46.15
SD	19.18	35.20
SY	**35.33**	**51.10**
TN	27.62	42.34
YE	22.12	38.96
Average	**26.48**	**42.30**

Table 5: Official spBLEU and chrF++ scores of the team's best submission for each dialect. The highest dialect is **bolded** while the lowest is underlined.
(EG: Egyptian, JO: Jordanian, LB: Lebanese, LY: Libyan, MA: Moroccan, MR: Mauritanian, OM: Omani, PS: Palestinian, SA: Saudi, SD: Sudanese, SY: Syrian, TN: Tunisian, and YE: Yemeni.)

the observed variation across dialects, we examined dialect-specific patterns that deviate from the overall trend. The four largest positive residuals cluster almost entirely around Levantine Arabic (Syrian, Jordanian, Palestinian, Lebanese) and Egyptian, dialects that are typologically close to one another and comparatively well-attested in general-purpose Arabic web text. We interpret this as evidence of positive cross-dialect transfer, with Syrian and Jordanian surpassing Lebanese and Palestinian, as these varieties benefit from shared representations learned across the Levantine mix rather than relying solely on individual training size.

Mauritanian performs worst despite moderate training data, trailing even zero-shot Libyan and Sudanese. This likely reflects typological proximity: transitional Libyan (Essgaer et al., 2025) and MSA-adjacent Sudanese (Abdalla, 2025) benefit from cross-dialect and pretraining transfer unavailable to Mauritanian, which is widely described as linguistically isolated from the other dialects(Taine-Cheikh, 2007). We view this as our best-supported explanation, though it remains literature-based rather than directly measured on our data.

9 Conclusion

In this work, we fine-tuned LLMs with 5B parameters or less for context-aware dialectal Arabic machine translation on the Alexandria dataset, evaluating different context strategies and sizes. Mismatched training and inference strategies yielded the strongest performance, while shorter conversation histories generally improved translation quality, with the optimal stage for context optimization varying across models. Quality-aware decoding further improved translation quality, particularly with a larger pool size. Future work will explore more diverse training data, longer conversational contexts, and optimization of per-dialect translation accuracy.

Limitations

Our models were trained only on the domains provided by the shared task organizers, so their performance on domains with specialized terminologies remains untested. The training set was also limited in size, and we expect larger-scale data would have improved translation quality.

Moreover, due to time constraints, there was no qualitative or human analysis, thus, evaluation relied heavily on the average overall scores. While Section 8 examines per-dialect performance in relation to training data volume and linguistic proximity, our explanations remain literature-grounded rather than empirically verified on our own data, and per-dialect optimization of the training or decoding pipeline was not attempted.

Finally, we limit our experiments to models of 5B parameters or less, as required by the constrained track and we choose the *ideal* strategy from preliminary experiments on only the baseline model then generalize it to the others also due to compute and time constraints.

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A Alexandria Dataset

Figure 1 presents an example dialogue instance from the Alexandria Dataset. The sample illustrates



Figure 1: Sample dialogue from the Alexandria Dataset in the Agriculture and Farming domain, showing a conversation between a male wholesale buyer and a female wholesale seller in Egyptian Arabic (Cairene dialect) and its parallel English translation.

the structure of the dataset, which includes bilingual dialogue pairs in Egyptian Arabic (Cairene dialect) and English, along with associated metadata describing the conversation context. This metadata captures information such as speaker role, gender, dialogue direction, and turn order.

B Prompting

To prompt the LLMs to generate the translation of each conversational turn based on the preceding turns and other contextual metadata, the same context-level prompt from EL Mekki et al. (2026) was used. The instructions are found in Figure 2.

C Model Baselines

This section presents the results of evaluating our 11 models on both the **default** and **ideal** setups prior to training. While base models perform best under the default setup, instruct models show gains under the ideal setup. Scores are shown in Table 6.

D Hyperparameter Configuration

Table 7 details the configuration used to train the top five models as well as the configuration that yielded the highest scores used on `gemma-3-4B-it`.

	Default Setup		Ideal Setup	
Model	**spBLEU**	**chrF++**	**spBLEU**	**chrF++**
NileChat-3B-Base	**9.30**	**26.60**	7.21	24.39
NileChat-3B	**8.61**	**20.19**	8.34	19.62
Qwen3-4B-Base	**5.55**	**17.47**	4.63	14.67
Qwen3-4B	4.29	17.72	**4.60**	**18.28**
Llama-3.2-3B	**1.76**	**6.76**	1.22	5.02
Llama-3.2-3B-Instruct	1.17	**8.71**	**1.20**	8.45
Tiny-aya-base	**4.72**	**11.85**	4.14	10.92
Tiny-aya-earth	2.46	**14.55**	**2.55**	14.48
Gemma-3-4B-it	3.30	**18.35**	**3.32**	18.18
Gemma-4-E4B-it	**0.27**	1.88	0.26	1.81
SILMA-Kashif-2B	1.04	9.85	**1.29**	**11.36**

Table 6: Evaluation results of base and instruct models across Default and Ideal prompt setups. The higher scoring setup for each model is bolded.

Context-Level Prompt Configuration

```
Translate the given turn of a conversation from English to <DIALECT>, considering the previous context if provided.

Input: { "country": "<Country>", "domain": "<Domain>", "participants": [...], "context": [<Previous_Turn_1>, <Previous_Turn_2>], "current_turn": { "speaker": "...", "text": "..." } }

Guidelines:
• Return the result strictly in valid JSON.
• Translate to <DIALECT> using Arabic script.
• Do not add any code, explanations, comments, or any other extra text.
• Keep the meaning and tone and respect the gender direction.
• Consider the country, the domain, the participants, and the speaker in your translation.
• Only translate the "text" field of the "current_turn".
• If a context is provided, do not translate it, and use it to inform your translation.

Output scheme: { "translation": "translation of the text from the current turn" }
```

Figure 2: The prompt configuration used for the English → Arabic Dialect evaluation.

Parameter	Original Config.	Best Config.
Epochs	1	2
Batch size	32	16
Grad. accum.	4	2
Eff. batch size	128	32
Learning rate	2e-4	2e-4
Weight decay	0	0.05
4-bit quant.	Yes	No
LoRA R / α	16/32	32/64
DoRA	No	Yes
Chat template	No	Yes
Max grad norm	1	1

Table 7: Training configurations tested during hyperparameter tuning experiments. Original Config. denotes the parameters of the configuration used for all five model, while Best Config. denotes the best set of hyperparameters tested later on `gemma-3-4b-it`.